

Statistical Modelling Exam preparation

January 11, 2024

Exercise 1

The `mtcars` dataset comprises fuel consumption (`mpg`: Miles/(US) gallon) and 8 aspects of automobile design and performance for 32 automobiles. Specifically, the covariates are

- `wt`: Weight (1000 lbs)
- `am`: Transmission (0 = automatic, 1 = manual)
- `cyl`: Number of cylinders
- `disp`: Displacement (cu.in.)
- `hp`: Gross horsepower
- `drat`: Rear axle ratio
- `qsec`: 1/4 mile time
- `vs`: Engine (0 = V-shaped, 1 = straight)

Fitting a Gaussian linear model in R produces the following output

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	15.5731	16.3817	0.951	0.3517
wt	-3.9437	1.2874	-3.063	0.0055
am = 1	2.7937	1.8682	1.495	0.1484
cyl	-0.2786	0.9348	-0.298	0.7683
disp	0.0147	0.0120	1.223	0.2338
hp	-0.0214	0.0162	??	0.1995
drat	0.8151	1.5101	0.540	0.5946
qsec	??	0.6587	1.229	0.2314
vs = 1	0.3684	2.0116	0.183	??

Residual standard error $\sqrt{\sum_{i=1}^{32} (y_i - \hat{y}_i)^2 / 23} = 2.544$

Coefficient $R^2 = 0.8678$

- Write the statistical model corresponding to the analysis (quantities and assumptions). Denote this model as “model A”.
- Write the parameter space and sample space.
- Complete the missing values in the table.
For “Pr(>|t|)” of `vs1`, write the meaning of the missing value and how to obtain it.
- Perform a test of overall significance of the model using a 5% significance level.

- e) On the same dataset, it is then estimated a reduced model (“model B”) that only includes the variables **wt** and **am**. The software estimates the following quantities:
 Residual standard error = 3.098
 Coefficient $R^2 = 0.7528$
 What procedure would you use to compare model A and model B? Following your chosen procedure, which model do you prefer?
- f) Starting from model B, it is then introduced as additional covariate the interaction between **wt** and **am**. Explain the resulting model and how you interpret the parameters.

Exercise 2

Given a set of $n = 30$ observations, consider fitting the model $Y_i \sim \text{Bernoulli}(\pi_i)$ where $\text{logit}(\pi_i) = \beta_1 + \beta_2 x_i$, with x_i is a dummy variable taking value 1 for the first 10 observations and 0 otherwise. Fitting this model returns the following output

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.3863	0.5590	2.480	0.01314
x	-2.0794	0.7826	-2.657	0.00788
Null deviance	47.111			
Residual deviance	39.112			

- a) Write the likelihood, log-likelihood and score functions for (β_1, β_2) . Write the fitted model.
- b) Compute the estimate of the probability $\hat{\pi}$ for $x = 0$ and $x = 1$. Obtain the odds for $x = 0$ and $x = 1$ and interpret them. Give an estimate of the odds ratio and interpret it.
- c) Test the hypothesis $H_0 : \beta_2 = -1$ vs $H_1 : \beta_2 < -1$.
- d) What are the two quantities “Null deviance” and “Residual deviance”?

		p						
		0.90	0.95	0.975	0.99	0.995	0.9975	0.999
standard Normal	z_p	1.2816	1.6449	1.9600	2.3263	2.5758	2.8070	3.0902
t with 21 df	$t_{21,p}$	1.3232	1.7207	2.0796	2.5176	2.8314	3.1352	3.5272
t with 22 df	$t_{22,p}$	1.3212	1.7171	2.0739	2.5083	2.8188	3.1188	3.5050
t with 23 df	$t_{23,p}$	1.3195	1.7139	2.0687	2.4999	2.8073	3.1040	3.4850
t with 31 df	$t_{31,p}$	1.3095	1.6955	2.0395	2.4528	2.7440	3.0221	3.3749
t with 32 df	$t_{32,p}$	1.3086	1.6939	2.0369	2.4487	2.7385	3.0149	3.3653
t with 33 df	$t_{33,p}$	1.3077	1.6924	2.0345	2.4448	2.7333	3.0082	3.3563

Table 1: Some quantiles of Gaussian and Student's t distribution: $p = \mathbb{P}(X \leq q_p)$. Columns correspond to probabilities p . Rows correspond to different distributions, in particular, for the t , each row corresponds to different degrees of freedom (df).

		p						
		0.90	0.95	0.975	0.99	0.995	0.9975	0.999
$f_{6,23;p}$		2.0472	2.5277	3.0232	3.7102	4.2591	4.8366	5.6486
$f_{7,23;p}$		1.9949	2.4422	2.9023	3.5390	4.0469	4.5807	5.3308
$f_{8,23;p}$		1.9531	2.3748	2.8077	3.4057	3.8822	4.3826	5.0853
$f_{9,23;p}$		1.9189	2.3201	2.7313	3.2986	3.7502	4.2243	4.8896
$f_{6,32;p}$		1.9668	2.3991	2.8356	3.4269	3.8886	4.3653	5.0211
$f_{7,32;p}$		1.9132	2.3127	2.7150	3.2583	3.6819	4.1185	4.7186
$f_{8,32;p}$		1.8702	2.2444	2.6202	3.1267	3.5210	3.9271	4.4846
$f_{9,32;p}$		1.8348	2.1888	2.5434	3.0208	3.3919	3.7738	4.2977
$f_{23,6;p}$		2.8223	3.8486	5.1284	7.3309	9.4992	12.2271	16.9460
$f_{23,7;p}$		2.5796	3.4179	4.4263	6.0921	7.6688	9.5865	12.7758
$f_{23,8;p}$		2.4086	3.1229	3.9587	5.2967	6.5260	7.9832	10.3357
$f_{23,9;p}$		2.2816	2.9084	3.6257	4.7463	5.7516	6.9197	8.7618
$f_{32,6;p}$		2.7953	3.7998	5.0521	7.2073	9.3290	11.9983	16.6155
$f_{32,7;p}$		2.5504	3.3670	4.3491	5.9712	7.5066	9.3740	12.4795
$f_{32,8;p}$		2.3777	3.0703	3.8806	5.1776	6.3691	7.7816	10.0616
$f_{32,9;p}$		2.2491	2.8543	3.5468	4.6282	5.5984	6.7255	8.5031

Table 2: Some quantiles of the F distribution: $p = \mathbb{P}(X \leq f_{df_1, df_2; p})$. Columns correspond to probabilities p . Rows correspond to different distributions, in particular, each row corresponds to different degrees of freedom (df).

EXERCISE 1

n = 32

$$Y_i = \beta_1 + \beta_2 x_{i1} + \beta_3 D_{i2} + \beta_4 x_{i3} + \beta_5 x_{i4} + \beta_6 x_{i5} + \beta_7 x_{i6} + \beta_8 x_{i7} + \beta_9 D_{i8}$$

Annotations:
 - x_{i1} : WEIGHT
 - D_{i2} : TRANSMISSION (0: AUTOMATIC, 1: MANUAL)
 - x_{i3} : DISPLACEMENT
 - x_{i4} : # CYLINDERS
 - x_{i5} : HP
 - x_{i6} : 1/16 MILE TIME
 - x_{i7} : REAR AXLE RATION
 - D_{i8} : ENGINE (0: V-SHAPED, 1: STRAIGHT)

ASSUMPTIONS

- $E(\epsilon_i) = 0 \rightarrow E(Y_i) = \beta_1 + \beta_2 \dots$
- $VAR(\epsilon_i) = \sigma^2 > 0$
- $Cov(\epsilon_i; \epsilon_k) = 0$ for $i \neq k \rightarrow Cov(Y_i; Y_k) = 0$
- CONDITIONAL $X_i = x_i$; $X_i = x_i$
- $\epsilon_i \sim N(0, \sigma^2)$ $i = 1, \dots, n$
- VECTORS $x_i \in R^p$ $i = 1, \dots, n$ LINEARLY INDEP. $\rightarrow RANK(X) = p$ FULL RANK

- SAMPLE SPACE R^n
- PARAM. SPACE $R^9 \cdot R^1$

	Estimate	Std. Error	t value	Pr(> t)
β_1 (Intercept)	15.5731	16.3817	0.951	0.3517
β_2 wt	-3.9437	1.2874	-3.063	0.0055
β_3 am = 1	2.7937	1.8682	1.495	0.1484
β_4 cyl	-0.2786	0.9348	-0.298	0.7683
β_5 disp	0.0147	0.0120	1.223	0.2338
β_6 hp	-0.0214	0.0162	2.000	0.1995
β_7 drat	0.8151	1.5101	0.540	0.5946
β_8 qsec	1.0000	0.6587	1.229	0.2314
β_9 vs = 1	0.3684	2.0116	0.183	0.8567

$$\sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{23}} = 2.544$$

$$R^2 = 0.3678$$

1 $\begin{cases} H_0: \beta_3 = 0 \\ H_1: \beta_3 \neq 0 \end{cases} \quad t_3^{OBS} = \frac{\hat{\beta}_3 - \beta_3}{SE(\hat{\beta}_3)} \rightarrow \hat{\beta}_3 = t_3^{OBS} \cdot SE(\hat{\beta}_3) = 0.309$

2 $\begin{cases} H_0: \beta_6 = 0 \\ H_1: \beta_6 \neq 0 \end{cases} \quad t_6^{OBS} = \frac{\hat{\beta}_6 - \beta_6}{SE(\hat{\beta}_6)} = -1.320$

3 $\begin{cases} H_0: \beta_9 = 0 \\ H_1: \beta_9 \neq 0 \end{cases} \quad \alpha^{OBS} = P(|t_{n-p}| \geq |t_{n-p}^{OBS}|) = 2 \cdot \min \{ P(t_{n-p} < -t_{n-p}^{OBS}), P(t_{n-p} > t_{n-p}^{OBS}) \} \approx 2(1 - 0.90) = 0.4$
 $n - p = 32 - 9 = 23 \quad t_{n-p}^{OBS} = 0.183$

4 $\alpha = 0.05$
 $\begin{cases} H_0: \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_6 = \beta_7 = \beta_8 = \beta_9 = 0 \\ H_1: \neg H_0 \end{cases}$

$$F_{p-p_0; n-p} \stackrel{H_0}{\sim} F_{OBS} = \frac{\frac{SSR}{p-1}}{\frac{SSE}{n-p}} = \frac{\frac{977.125}{9-1}}{\frac{142.8}{32-9}} = 12.379$$

$$= \frac{R^2}{1-R^2} \frac{n-p}{p-1} = 12.373$$

$$\alpha^{OBS} = P(|F_{p-p_0; n-p}| \geq F_{OBS}) \approx (1 - 0.999) \approx 0$$

Since $\alpha^{OBS} < \alpha = 0.05$ | RES. H_0

5 Model B: $Y_i = \beta_1 + \beta_2 x_{i1} + \beta_3 D_{i2}$

- $SSE(B) = 3.098$
- $R^2(B) = 0.7528$

$$\begin{cases} H_0: \beta_4 = \beta_5 = \beta_6 = \beta_7 = \beta_8 = \beta_9 = 0 \\ H_1: \neg H_0 \end{cases} \quad F_{p-p_0; n-p} \stackrel{H_0}{\sim} F_{OBS} = \frac{\frac{SSE(B) - SSE(A)}{p-p_0}}{\frac{SSE(A)}{n-p}} = \frac{\frac{3.098 - 142.8}{9-3}}{\frac{142.8}{32-9}} = -3.75$$

$$\alpha^{OBS} = P(|F_{p-p_0; n-p}| \geq F_{OBS}) \approx (1 - 0.999) \approx 0$$

Since $\alpha^{OBS} < \alpha$ | RES. H_0
 PREFER LARGEST MODEL

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mod B: $Y_i = \beta_1 + \beta_2 x_{1i} + \beta_3 D_{2i} + \beta_{40} x_{1i} D_{2i}$

IF AUTOMATIC

$$Y_i = \beta_1 + \beta_2 x_{1i}$$

KEEPING EVERYTHING CONSTANT, ON AVERAGE, IF A CAR IS AUTOMATIC THE FUEL CONSUMPTION WILL DECREASE BY -3.94 IF WE INCREASE THE WEIGHT BY 1000 LBS

IF MANUAL

$$Y_i = (\beta_1 + \beta_3) + (\beta_2 + \beta_{40}) x_{1i}$$

KEEPING EVERYTHING CONSTANT, ON AVERAGE, IF A CAR IS MANUAL, THE FUEL CONSUMPTION WILL INCREASE BY $(\beta_2 + \beta_{40})$ IF WE INCREASE THE WEIGHT BY 1000 LBS

Exercise 2

Given a set of $n = 30$ observations, consider fitting the model $Y_i \sim \text{Bernoulli}(\pi_i)$ where $\text{logit}(\pi_i) = \beta_1 + \beta_2 x_i$, with x_i is a dummy variable taking value 1 for the first 10 observations and 0 otherwise. Fitting this model returns the following output

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- Test the hypothesis $H_0 : \beta_2 = -1$ vs $H_1 : \beta_2 < -1$.
- What are the two quantities "Null deviance" and "Residual deviance"?

$n=30$

$$\eta_i = \beta_1 + \beta_2 x_i$$

$$Y_i \sim \text{BERNOULLI}(\pi_i)$$

$$g(\pi_i) = \text{LOG}\left(\frac{\pi_i}{1-\pi_i}\right) = \eta_i$$

$$\text{LOGIT}(\pi_i) = \beta_1 + \beta_2 x_i$$

$$\begin{cases} 1 & \text{if } i=1, \dots, 10 \\ 0 & \text{OTHERWISE} \end{cases}$$

BERNOULLI

$$f(Y_i; \theta) = \pi_i^{y_i} (1-\pi_i)^{1-y_i}$$

$$\text{LOG}\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_1 + \beta_2 x_i \rightarrow \pi_i = \frac{e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}} = \frac{e^{\tilde{x}_i^T \theta}}{1 + e^{\tilde{x}_i^T \theta}}$$

LIKELIHOOD

$$L(\theta) = \prod (\pi_i^{y_i} (1-\pi_i)^{1-y_i}) = \prod \left(\frac{e^{\tilde{x}_i^T \theta}}{1 + e^{\tilde{x}_i^T \theta}} \right)^{y_i} \prod \left(1 - \frac{e^{\tilde{x}_i^T \theta}}{1 + e^{\tilde{x}_i^T \theta}} \right)^{1-y_i} = \prod \left(\frac{e^{\tilde{x}_i^T \theta}}{1 + e^{\tilde{x}_i^T \theta}} \right)^{y_i} \prod \left(\frac{1}{1 + e^{\tilde{x}_i^T \theta}} \right)^{1-y_i}$$

LOG-LIKELIHOOD

$$\begin{aligned} \ell(\theta) = \text{LOG}(L(\theta)) &= \text{LOG} \left(\prod \left(\frac{e^{\tilde{x}_i^T \theta}}{1 + e^{\tilde{x}_i^T \theta}} \right)^{y_i} \prod \left(\frac{1}{1 + e^{\tilde{x}_i^T \theta}} \right)^{1-y_i} \right) = \text{LOG} \left(\prod \frac{e^{\tilde{x}_i^T \theta}}{1 + e^{\tilde{x}_i^T \theta}} \right)^{y_i} + \text{LOG} \left(\prod \frac{1}{1 + e^{\tilde{x}_i^T \theta}} \right)^{1-y_i} \\ &= \sum y_i \text{LOG} \left(\frac{e^{\tilde{x}_i^T \theta}}{1 + e^{\tilde{x}_i^T \theta}} \right) + \sum (1-y_i) \text{LOG} \left(\frac{1}{1 + e^{\tilde{x}_i^T \theta}} \right) = \sum y_i \left[\text{LOG}(e^{\tilde{x}_i^T \theta}) - \text{LOG}(1 + e^{\tilde{x}_i^T \theta}) \right] + \sum (1-y_i) \left[\cancel{\text{LOG}(1)} - \text{LOG}(1 + e^{\tilde{x}_i^T \theta}) \right] \\ &= \sum y_i \left[\tilde{x}_i^T \theta - \text{LOG}(1 + e^{\tilde{x}_i^T \theta}) \right] - \sum (1-y_i) \text{LOG}(1 + e^{\tilde{x}_i^T \theta}) \\ &= \sum y_i \tilde{x}_i^T \theta - \cancel{\sum y_i \text{LOG}(1 + e^{\tilde{x}_i^T \theta})} - \sum 1 \text{LOG}(1 + e^{\tilde{x}_i^T \theta}) + \cancel{\sum y_i \text{LOG}(1 + e^{\tilde{x}_i^T \theta})} \\ &= \sum y_i \tilde{x}_i^T \theta - \sum \text{LOG}(1 + e^{\tilde{x}_i^T \theta}) \end{aligned}$$

SCORE FUNCTION

$$l(\theta) = \sum y_i \tilde{x}_i^T \theta - \sum \log(1 + e^{\tilde{x}_i^T \theta}) = \sum y_i (\beta_1 + \beta_2 x_i) - \sum \log(1 + e^{\beta_1 + \beta_2 x_i})$$

$$\begin{cases} a = \frac{d}{d\beta_1} l(\theta) \\ b = \frac{d}{d\beta_2} l(\theta) \end{cases}$$

$$a = \frac{d}{d\beta_1} \left[\sum y_i (\beta_1 + \beta_2 x_i) - \sum \log(1 + e^{\beta_1 + \beta_2 x_i}) \right] = \frac{d}{d\beta_1} \sum y_i \beta_1 - \frac{d}{d\beta_1} \sum \log(1 + e^{\beta_1 + \beta_2 x_i}) = \sum y_i - \sum \frac{e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}}$$

$$b = \frac{d}{d\beta_2} \left[\sum y_i (\beta_1 + \beta_2 x_i) - \sum \log(1 + e^{\beta_1 + \beta_2 x_i}) \right] = \sum y_i x_i - \sum \frac{x_i e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}}$$

THE FITTED MODEL IS: $Y_i \sim \text{BER}(\pi_i)$ $\text{LOGIT}(\hat{\pi}_i) = 1.3863 + 2.0794x_i$

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$x_i = 0$

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_1 + \beta_2 x_i \quad \therefore \pi_i = \frac{e^{\beta_1 + \beta_2 x_i}}{1 + e^{\beta_1 + \beta_2 x_i}}$$

$x_i = 0$

$$\bullet P(y_i = 1 | x_i = 0) = \frac{e^{\beta_1}}{1 + e^{\beta_1}}$$

$$\bullet P(y_i = 0 | x_i = 0) = 1 - \frac{e^{\beta_1}}{1 + e^{\beta_1}} = \frac{1}{1 + e^{\beta_1}}$$

$$\bullet P(y_i = 1 | x_i = 1) = \frac{e^{\beta_1 + \beta_2}}{1 + e^{\beta_1 + \beta_2}}$$

$$\bullet P(y_i = 0 | x_i = 1) = 1 - \frac{e^{\beta_1 + \beta_2}}{1 + e^{\beta_1 + \beta_2}} = \frac{1}{1 + e^{\beta_1 + \beta_2}}$$

ODDS-SCALE

$$\bullet \frac{P(y_i = 1 | x_i = 0)}{P(y_i = 0 | x_i = 0)} = \frac{e^{\beta_1}}{1 + e^{\beta_1}} \cdot \frac{1 + e^{\beta_1}}{e^{\beta_1}} = e^{\beta_1}$$

$e^{\hat{\beta}_1} = 4$ WHEN $x_i = 0$ I EXPECT 400 SUCCESS EVERY 100 FAILURE

$e^{\hat{\beta}_1} = 0.12$ THE ODDS FOR THE GROUP $x_i = 0$ ARE MULTIPLIED BY 0.125 TO OBTAIN THE ODDS AT $x_i = 1$

$$\text{ODDS-RATIO} = \frac{e^{\beta_1 + \beta_2}}{e^{\beta_1}} = e^{\beta_2}$$

$$\bullet \frac{P(y_i = 1 | x_i = 1)}{P(y_i = 0 | x_i = 1)} = \frac{e^{\beta_1 + \beta_2}}{1 + e^{\beta_1 + \beta_2}} \cdot \frac{1 + e^{\beta_1 + \beta_2}}{e^{\beta_1 + \beta_2}} = e^{\beta_1 + \beta_2}$$

$e^{\hat{\beta}_1 + \hat{\beta}_2} = 0.5$ WHEN $x_i = 1$ I EXPECT 50 SUCCESS EVERY 100 FAILURES

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$$\begin{cases} H_0: \beta_2 = -1 \\ H_1: \beta_2 < -1 \end{cases}$$

$$Z_2 \sim N(0,1) \xrightarrow{H_0} Z_2^{\text{OBS}} = \frac{\hat{\beta}_2 - \beta_2}{\sqrt{\hat{\sigma}(\hat{\beta})^{-1}_{22}}} = -1.37$$

$$\alpha^{\text{OBS}} = P(|Z_2| \geq |Z_2^{\text{OBS}}|) = 2(1 - \Phi(|Z_2^{\text{OBS}}|))$$

$$= 1 - 0.92 = 0.08$$



• WHEN $\alpha = 0.1$ I REJECT H_0 SINCE $\alpha^{\text{OBS}} < \alpha$

• WHEN $\alpha = 0.05$ I NOT REJECT H_0 SINCE $\alpha^{\text{OBS}} > \alpha$
 $\hookrightarrow \text{AND } \alpha = 0.01$

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$$\bullet D(\text{MODEL}) = 2\{\tilde{l}(\text{SAT.}) - \tilde{l}(\text{MODEL})\} \leftarrow \text{MODEL DEVIANCE}$$

$$\bullet D(\text{MODEL}) = 2\{\tilde{l}(\text{SAT.}) - \hat{l}(\text{MODEL})\} \leftarrow \text{RESIDUAL DEVIANCE}$$